

Last Name:

First Name:

ID:

1. This test has 7 pages including the cover page and 8 questions
2. Do not separate the sheets (leave the staple in)
3. Print your name, student number and section number on the first page. Write your last name on each subsequent page.
4. Answer each question in the space provided. If you need more space, clearly indicate where the rest of your solution is to be found. **USE THE BACK OF THE SHEET FOR EXTRA CALCULATIONS.**
5. It is important to be specific about what you are calculating, otherwise it is impossible to award partial marks for solutions containing errors. Also, define all used events.
6. No programming calculators and cell phone calculators are allowed.

Question	Maximum mark	Actual mark
1	10	
2	15	
3	10	
4	5	
5	5	
6	5	
7	10	
8	6	
Total	66 pts	66 pts

Question 1.

10 pts

Suppose Y is a discrete random variable with probability distribution function

y	0	1	2	3	4
$g(y)$	b	$\frac{1}{4}$	$\frac{1}{8}$	$2b$	$\frac{1}{4}$

Compute the mean, variance, and standard deviation of Y . Your answers must be numeric (no symbols).

Solution 1.

- Determine b

Since the sum of all probabilities equals 1,

$$b + \frac{1}{4} + \frac{1}{8} + 2b + \frac{1}{4} = 1$$

Combine like terms:

$$3b + \frac{5}{8} = 1$$

$$3b = 1 - \frac{5}{8} = \frac{3}{8}$$

$$\boxed{b = \frac{1}{8}} \quad 2 \text{ pts}$$

- Compute the mean

$$E(Y) = \sum yg(y)$$

$$E(Y) = (0)(0.125) + (1)(0.25) + (2)(0.125) + (3)(0.25) + (4)(0.25)$$

$$E(Y) = 0 + 0.25 + 0.25 + 0.75 + 1.00 = 2.25$$

$$\boxed{E(Y) = 2.25} \quad 2 \text{ pts}$$

- Compute $E(Y^2)$

$$E(Y^2) = \sum y^2 g(y)$$

$$= (0)^2(0.125) + (1)^2(0.25) + (2)^2(0.125) + (3)^2(0.25) + (4)^2(0.25)$$

$$= 0 + 0.25 + 0.50 + 2.25 + 4.00 = 7.00$$

$$\boxed{E(Y^2) = 7.00} \quad 2 \text{ pts}$$

- Compute the variance

$$\begin{aligned}\text{Var}(Y) &= E(Y^2) - [E(Y)]^2 \\ \text{Var}(Y) &= 7.00 - (2.25)^2 = 7.00 - 5.0625 = 1.9375\end{aligned}$$

$$\boxed{\text{Var}(Y) = 1.9375} \quad 2 \text{ pts}$$

- Compute the standard deviation

$$\sigma_Y = \sqrt{\text{Var}(Y)} = \sqrt{1.9375} \approx 1.39$$

$$\boxed{\sigma_Y = 1.39} \quad 2 \text{ pts}$$

Question 2.

15 pts

Given the function

$$f(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) (3 pts) Show that f is a pdf of a random variable, denoted by X .
- (b) (4 pts) Find the cdf of X .
- (c) (4 pts) Calculate $\mathbb{E}(X)$.
- (d) (4 pts) Find the mgf $M_X(t)$

Solution 2.

(a) To show that f is a pdf, we check the two conditions:

- Nonnegativity:

$$f(x) = \frac{1}{2} e^{-x/2} \geq 0 \quad \text{for } x > 0, \quad f(x) = 0 \text{ otherwise.}$$

- Total integral equals 1:

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \frac{1}{2} e^{-x/2} dx = \frac{1}{2} \left[-2e^{-x/2} \right]_0^{\infty} = \frac{1}{2}(0 + 2) = 1.$$

Thus f is a valid pdf.

(b) The cdf is

$$F(x) = \int_{-\infty}^x f(t) dt.$$

If $x \leq 0$,

$$F(x) = 0.$$

If $x > 0$,

$$F(x) = \int_0^x \frac{1}{2} e^{-t/2} dt = \left[-e^{-t/2} \right]_0^x = 1 - e^{-x/2}.$$

Hence

$$F(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-x/2}, & x > 0. \end{cases}$$

(c) The $\mathbb{E}(X)$ can be calculated as

$$\mathbb{E}(X) = \int_0^\infty x \cdot \frac{1}{2} e^{-x/2} dx.$$

We use integration by parts with

$$u = x, \quad dv = \frac{1}{2} e^{-x/2} dx,$$

so

$$du = dx, \quad v = -e^{-x/2}.$$

Then

$$\mathbb{E}(X) = uv \Big|_0^\infty - \int_0^\infty v du = \left[-xe^{-x/2} \right]_0^\infty + \int_0^\infty e^{-x/2} dx.$$

Compute each term:

1. Boundary term:

$$\lim_{x \rightarrow \infty} (-xe^{-x/2}) = 0, \quad (-0 \cdot e^0) = 0,$$

so the whole boundary term equals 0.

2. Remaining integral:

$$\int_0^\infty e^{-x/2} dx = \left[-2e^{-x/2} \right]_0^\infty = 2.$$

Thus

$$\mathbb{E}(X) = 2.$$

(d) The mgf is

$$M_X(t) = \mathbb{E}(e^{tX}) = \int_0^\infty e^{tx} \cdot \frac{1}{2} e^{-x/2} dx = \frac{1}{2} \int_0^\infty e^{x(t-1/2)} dx.$$

For convergence we require $t < \frac{1}{2}$. Then

$$M_X(t) = \frac{1}{2} \left[\frac{1}{t-1/2} e^{x(t-1/2)} \right]_0^\infty = \frac{1}{2} \left(0 - \frac{1}{t-1/2} \right) = \frac{1}{1-2t}.$$

Thus,

$$M_X(t) = \frac{1}{1-2t}, \quad t < \frac{1}{2}.$$

Question 3.

10 pts

The probability mass function of a random variable X is given by

$$f(x) = \frac{k \cdot 4^x}{x!}, \quad x = 0, 1, 2, \dots$$

(a) (2 pts) find the constant k

(b) (4 pts) Find the mgf

(c) (4 pts) Compute $\mathbb{E}(X)$ and $\mathbb{Var}(X)$

Solution 3.

(a) Find k

$$\begin{aligned}
\sum_{x=0}^{\infty} P(X=x) = 1 &\Leftrightarrow \sum_{x=0}^{\infty} \frac{k \cdot 4^x}{x!} = 1 \\
&\Leftrightarrow k \underbrace{\sum_{x=0}^{\infty} \frac{4^x}{x!}}_{e^4} = 1 \\
&\Leftrightarrow k \cdot e^4 = 1 \\
&\Leftrightarrow k = \frac{1}{e^4} = e^{-4}
\end{aligned}$$

Thus, $P(X=x) = \frac{e^{-4} 4^x}{x!}$ (2 pts)

(b) The mgf can be derived as follow:

$$\begin{aligned}
M_X(t) = \mathbb{E}[e^{tX}] &= \sum_{x=0}^{\infty} e^{tx} P(X=x) \\
&= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-4} \cdot 4^x}{x!} \\
&= e^{-4} \sum_{x=0}^{\infty} \frac{(4e^t)^x}{x!} \\
&= e^{-4} e^{4e^t} \\
&= e^{4(e^t-1)}, \quad t \in \mathbb{R}. \quad (4 \text{ pts})
\end{aligned}$$

(c) This mgf of the Poisson distribution with parameter $\lambda = 4$, similar calculation as in lecture notes shows that

$$\mathbb{E}[X] = \mathbb{V}\text{ar}[X] = \lambda = 4 \quad (4 \text{ pts})$$

Question 4.

5 pts

A factory produces microchips, each defective with probability $p = 0.04$. A sample of $n = 25$ chips is tested.

- (a) (1 pts) State the distribution of X , the number of defective chips.
- (b) (2 pts) Compute $P(X \leq 1)$.
- (c) (2 pts) Compute $E[X]$ and $\text{Var}(X)$.

Solution 4.

(a) $X \sim \text{Bin}(25, 0.04)$ (1 pts).

(b)

$$P(X \leq 1) = \binom{25}{0} (0.96)^{25} + \binom{25}{1} (0.04)(0.96)^{24} \approx 0.73581 \quad (2 \text{ pts})$$

(c)

$$E[X] = np = 25(0.04) = 1, \quad (1 \text{ pts}) \quad \text{Var}(X) = np(1-p) = 25(0.04)(0.96) = 0.96. \quad (1 \text{ pts})$$

Question 5.

5 pts

A student guesses randomly on a multiple-choice test with 4 choices per question. Let X be the number of attempts needed to get 3 correct answers.

- (a) (1 pts) Identify the distribution of X .
- (b) (2 pts) Compute $P(X = 7)$.
- (c) (2 pts) Compute $E[X]$ and $\text{Var}(X)$.

Solution 5.

Probability of success: $p = 1/4$.

- (a) (1 pts) $X \sim \text{NegBin}(r = 3, p = 1/4)$.
- (b)

$$P(X = 7) = \binom{6}{2} (1/4)^3 (3/4)^4 \approx 0.075. \quad (2 \text{ pts})$$

- (c)

$$E[X] = \frac{r}{p} = \frac{3}{1/4} = 12, (1 \text{ pts}) \quad \text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{3(3/4)}{(1/4)^2} = 36. (1 \text{ pts})$$

Question 6.

5 pts

Among the 300 employees of a company, 240 are union members, whereas the others are not. If 6 of the employees are chosen by lot to serve on a committee that administers the pension fund, find the probability that 4 of the 6 will be union members using

- (a) (2.5 pts) the formula for the hypergeometric distribution;
- (b) (2.5 pts) the binomial distribution approximation

Solution 6.

- (a) Using the Hypergeometric distribution with $n = 300, M = 240, x = 4$, we get,

$$P(X = x) = \frac{\binom{240}{4} \binom{60}{2}}{\binom{300}{6}} \approx 0.2478. \quad (2.5 \text{ pts})$$

- (b) Now, using the binomial distribution with $n = 6, \theta = 0.8$, we get

$$P(X = x) = \binom{6}{4} (0.8)^4 (0.2)^2 \approx 0.24576. \quad (2.5 \text{ pts})$$

Question 7.

10 pts

An insurance company knows that the yearly total number of damages from natural disasters in a given locality is a Poisson random variable, $X \sim \text{Poi}(\lambda)$. They also know that the probability of no damages in given three years is 0.1.

- (a) (5 pts) Find the mean and variance of number of damages per year.
- (b) (5 pts) If the company pays 1000\$ for each damage in a year after the first two such damages in that year. What is the expected amount of money paid by the company in a year.

Solution 7.

- (a) (5 pts) First, we find λ , to this end let Y denotes the number of damages in three years. Then, $Y \sim \text{Poi}(3\lambda)$, it follows

$$\begin{aligned}
 P(Y = 0) = 0.1 &\iff \frac{e^{-3\lambda}(3\lambda)^0}{0!} = 0.1 \\
 &\iff e^{-3\lambda} = 0.1 \\
 &\iff -3\lambda = \ln(0.1) \\
 &\iff \lambda = -\frac{\ln(0.1)}{3} = 0.7675 \quad (3 \text{ pts})
 \end{aligned}$$

Finally, the mean and variance of Poisson distribution is the parameter it self, thus

$$\mathbb{E}[X] = \text{Var}(X) = \lambda = 0.7675. \quad (2 \text{ pts})$$

- (b) (5 pts) The expected amount of money by the company in a year is given by

$$\begin{aligned}
 \mathbb{E}[1000(X - 2)] &= 1000 \sum_{x=3}^{+\infty} (x - 2) \frac{\lambda^x e^{-\lambda}}{x!} \\
 &= 1000 \left[\sum_{x=0}^{+\infty} (x - 2) \frac{\lambda^x e^{-\lambda}}{x!} - \sum_{x=0}^2 (x - 2) \frac{\lambda^x e^{-\lambda}}{x!} \right] \\
 &= 1000 \left[\underbrace{\sum_{x=0}^{+\infty} x \frac{\lambda^x e^{-\lambda}}{x!}}_{=\lambda} - 2 \underbrace{\sum_{x=0}^{+\infty} \frac{\lambda^x e^{-\lambda}}{x!}}_{=1} - \sum_{x=0}^2 (x - 2) \frac{\lambda^x e^{-\lambda}}{x!} \right] \\
 &= 1000 [\lambda - 2 + 2e^{-\lambda} + \lambda e^{-\lambda}] \\
 &= 1000 [0.7675 + 2e^{-0.7675} + (0.7675) \cdot e^{-0.7675}] \\
 &= 52.1
 \end{aligned}$$

Question 8.**6 pts**

A factory has two machines. Machine A produces 60% of all items and has a defect probability of 4%. Machine B produces 40% of items and has a defect probability of 8%.

A random sample of $n = 20$ items is taken. Let X be the number of defective items.

- (a) (2 pts) Compute the defect probability $p = P(\text{defect})$.
- (b) (2 pts) Find the distribution of X .
- (c) (2 pts) Compute $P(X \geq 3)$.

Solution 8.

- (a)

$$p = 0.6(0.04) + 0.4(0.08) = 0.056. \quad (2 \text{ pts})$$

- (b)

$$X \sim \text{Bin}(n = 20, \theta = 0.056). \quad (2 \text{ pts})$$

(c)

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - \sum_{k=0}^2 \binom{20}{k} (0.056)^k (0.944)^{20-k} \approx 0.0983 \quad (2 \text{ pts})$$